



chapter6b_mgfs (1)

Chapter 6B: Moment Generating Functions

MATH/STAT 304: Probability I

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Learning Goals

By the end of this lecture, students should be able to:

- define the moment generating function (MGF),
- compute MGFs of common distributions,
- extract moments from MGFs,
- understand how MGFs determine distributions,
- compute distributions of sums using MGFs,
- derive moments of Exponential and Normal distributions using MGFs.

Definition of the MGF

Definition

The moment generating function (MGF) of X is

$$M_X(t) = \mathbb{E}(e^{tX})$$

provided the expectation is finite in some open interval around 0. Equivalently,

$$M_X(t) = \begin{cases} \sum_{k=0}^{\infty} e^{t^k} p_X(k), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} e^{tx} f_X(x) dx, & \text{if } X \text{ is continuous.} \end{cases}$$

Quick check

Every valid MGF satisfies

$$M_X(0) = 1.$$

MGFs: Discrete Distributions I

Binomial

If $X \sim \text{Bin}(n, p)$, then

$$M_X(t) = e^{t^2} (1-p)^{n-1} + e^{t^2} (1-p)^{n-1} = (pe^t + q)^n$$

Discrete Uniform

If $X \sim \text{DisUnif}\{1, \dots, m\}$, then

$$M_X(t) = \frac{1}{m} \sum_{k=1}^m e^{tk} = \frac{1}{m} (e^t + e^{2t} + \dots + e^{mt}) = \frac{e^t(e^{mt} - 1)}{m(e^t - 1)}$$

MGFs: Discrete Distributions II

Binomial

If $X \sim \text{Bin}(n, p)$, then

$$M_X(t) = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} e^{tk} = \sum_{k=0}^n \binom{n}{k} (pe^t)^k q^{n-k} = (q + pe^t)^n$$

Poisson

If $X \sim \text{Pois}(\lambda)$, then

$$M_X(t) = \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} e^{tk} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}$$

MGFs: Discrete Distributions III

Geometric

If $X \sim \text{Geom}(p)$, where X counts failures before the first success, then

$$M_X(t) = \sum_{k=0}^{\infty} p(1-p)^k e^{tk} = p \sum_{k=0}^{\infty} (pe^t)^k = \frac{p}{1 - pe^t}, \quad pe^t < 1$$

Negative Binomial

If $X \sim \text{NegBin}(r, p)$, where X counts failures before the r th success, then

$$M_X(t) = \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} p^r q^k e^{tk} = p^r \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} (pe^t)^k q^{r-k} = \left(\frac{p}{1 - pe^t} \right)^r$$

MGFs: Discrete Distributions IV

Hypergeometric

If

$$X \sim \text{HGeom}(n, K, n)$$

then

$$\mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{n-K}{n-k}}{\binom{n}{n}}$$

where

$$\max(0, n-K) \leq k \leq \min(K, n)$$

Therefore,

$$M_X(t) = \sum_{k=0}^n \frac{\binom{K}{k} \binom{n-K}{n-k}}{\binom{n}{n}} e^{tk} = \frac{\binom{K}{0} \binom{n-K}{n} + \dots + \binom{K}{n} \binom{n-K}{0}}{\binom{n}{n}}$$

Note: Unlike Binomial or Poisson, this MGF usually does not simplify to a nice elementary closed form.

MGFs: Continuous Distributions I

Uniform

If $X \sim \text{Unif}(a, b)$, then

$$M_X(t) = \int_a^b e^{tx} \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{e^{bx} - e^{ax}}{t} \right]_a^b = \frac{e^{bt} - e^{at}}{t(b-a)}, \quad t \neq 0$$

Also, $M_X(0) = 1$.

MGFs: Continuous Distributions II

Exponential

If $X \sim \text{Exp}(\lambda)$, then

$$M_X(t) = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} e^{(t-\lambda)x} dx = \frac{\lambda}{\lambda - t}, \quad t < \lambda$$

Gamma

If $X \sim \text{Gamma}(\alpha, \lambda)$ with shape α and rate λ , then

$$M_X(t) = \int_0^{\infty} e^{tx} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^{\infty} x^{\alpha-1} e^{-(\lambda-t)x} dx$$

Let $u = (\lambda - t)x$. Then $x = u/(\lambda - t)$, $dx = du/(\lambda - t)$, and

$$M_X(t) = \frac{\lambda^\alpha}{\Gamma(\alpha) (\lambda - t)^\alpha} \int_0^{\infty} u^{\alpha-1} e^{-u} du = \frac{\lambda^\alpha}{(\lambda - t)^\alpha} = \left(\frac{\lambda}{\lambda - t} \right)^\alpha, \quad t < \lambda$$

MGF of $N(0, 1)$

Standard Normal

Let $Z \sim N(0, 1)$. Then

$$M_Z(t) = \mathbb{E}(e^{tZ}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-x^2/2} dx$$

Complete the square:

$$tx - \frac{x^2}{2} = -\frac{1}{2}(x^2 - 2tx) = -\frac{1}{2}(x - t)^2 + \frac{t^2}{2}$$

Therefore,

$$M_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} e^{t^2/2} dx = e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} dx$$

Since the last integral equals $\sqrt{2\pi}$,

$$M_Z(t) = e^{t^2/2}$$

MGF of $N(\mu, \sigma^2)$

Normal

Let $X \sim N(\mu, \sigma^2)$. Write

$$X = \mu + \sigma Z, \quad Z \sim N(0, 1)$$

Then

$$M_X(t) = \mathbb{E}(e^{tX}) = \mathbb{E}(e^{t(\mu + \sigma Z)}) = e^{t\mu} \mathbb{E}(e^{t\sigma Z})$$

Using

$$M_Z(t) = e^{t^2/2}$$

with $s = t\sigma$, we get

$$M_X(t) = e^{t\mu} M_Z(t\sigma) = e^{t\mu} e^{(t\sigma)^2/2} = e^{t\mu + \frac{1}{2}\sigma^2 t^2}$$

Moments from MGFs

MGFs Generate Moments

Theorem

The n th population moment is obtained by differentiating the MGF:

$$\mathbb{E}(X^n) = M_X^{(n)}(0)$$

Examples:

$$\mathbb{E}(X) = M_X'(0), \quad \mathbb{E}(X^2) = M_X''(0)$$

Proof: MGFs Generate Moments

Recall that the moment generating function is

$$M_X(t) = \mathbb{E}(e^{tX})$$

Differentiate once:

$$M_X'(t) = \frac{d}{dt} \mathbb{E}(e^{tX}) = \mathbb{E} \left(\frac{d}{dt} e^{tX} \right) = \mathbb{E}(X e^{tX})$$

Now plugging in $t = 0$,

$$M_X'(0) = \mathbb{E}(X e^{0X}) = \mathbb{E}(X)$$

Differentiate another time noting that X is a constant w.r.t. t :

$$M_X''(t) = \frac{d}{dt} \mathbb{E}(X e^{tX}) = \mathbb{E} \left(\frac{d}{dt} X e^{tX} \right) = \mathbb{E}(X^2 e^{tX})$$

Plugging in $t = 0$,

$$M_X''(0) = \mathbb{E}(X^2 e^{0X}) = \mathbb{E}(X^2)$$

Similarly, differentiating n times gives

$$M_X^{(n)}(t) = \mathbb{E}(X^n e^{tX})$$

Therefore, at $t = 0$,

$$M_X^{(n)}(0) = \mathbb{E}(X^n e^{0X}) = \mathbb{E}(X^n)$$

Taylor Expansion Around a Point

Let f be infinitely differentiable in a neighborhood of a . Its n th-order Taylor polynomial around a is

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$

Taylor's theorem gives

$$f(x) = T_n(x) + R_n(x)$$

where, under the usual differentiability assumptions, the remainder can be written as

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}$$

for some ξ between a and x .

Infinite Taylor Series

If $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$, then

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k$$

Taylor Expansion of the Exponential Function

For

$$f(x) = e^x$$

we have

$$f^{(k)}(a) = e^a$$

Expanding around $a = 0$,

$$T_n(x) = \sum_{k=0}^n \frac{e^0}{k!} x^k = \sum_{k=0}^n \frac{x^k}{k!}$$

so

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

The exponential function is analytic on all of $\mathbb{R}$, or this series converges to e^x for every $x \in \mathbb{R}$. Now substitute

$$x = tX$$

for every fixed realization of X :

$$e^{tX} = \sum_{k=0}^{\infty} \frac{(tX)^k}{k!} = 1 + tX + \frac{t^2 X^2}{2!} + \dots$$

Thus, the identity holds pointwise for every value of X .

Taking Expectations: Why Is It Valid?

The moment-generating function of X is

$$M_X(t) = \mathbb{E}(e^{tX})$$

provided this expectation is finite. Using the Taylor expansion,

$$M_X(t) = \mathbb{E} \left(\sum_{k=0}^{\infty} \frac{t^k X^k}{k!} \right) = \sum_{k=0}^{\infty} \frac{t^k}{k!} \mathbb{E}(X^k)$$

To write

$$\mathbb{E} \left(\sum_{k=0}^{\infty} \frac{t^k X^k}{k!} \right) = \sum_{k=0}^{\infty} \frac{t^k}{k!} \mathbb{E}(X^k)$$

we must justify interchanging the expectation and the infinite sum. A sufficient condition is that, for some $r > 0$,

$$\mathbb{E}(e^{r|X|}) < \infty$$

Moments as Taylor Coefficients of the MGF

Therefore,

$$M_X(t) = \mathbb{E} \left(\sum_{k=0}^{\infty} \frac{t^k X^k}{k!} \right) = \sum_{k=0}^{\infty} \frac{t^k}{k!} \mathbb{E}(X^k)$$

whenever the required interchange is valid. Equivalently,

$$M_X(t) = 1 + t\mathbb{E}(X) + \frac{t^2}{2!} \mathbb{E}(X^2) + \dots$$

Comparing this with the Taylor expansion

$$M_X(t) = \sum_{k=0}^{\infty} \frac{M_X^{(k)}(0)}{k!} t^k$$

we obtain

$$M_X^{(k)}(0) = \mathbb{E}(X^k)$$

whenever M_X exists in an open neighborhood of 0.

Main Idea

The k th raw moment of X is the k th derivative of its MGF evaluated at $t = 0$.

Example: Bernoulli Moments

$X \sim \text{Bern}(p)$

For

$$M_X(t) = pe^t + q$$

we compute:

$$M_X'(t) = pe^t \Rightarrow M_X'(0) = p$$

Thus:

$$\mathbb{E}(X) = p$$

So

$$\mathbb{E}(X^2) = M_X''(0) = p$$

$\Rightarrow \text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = p - p^2 = p(1-p)$

MGF Uniqueness Theorem

Theorem (Uniqueness of MGF)

If two random variables have the same MGF on an interval around 0, then they have the same distribution, i.e., the moment generating function uniquely determines the distribution.

Thus MGFs characterize distributions, similarly to PDFs and CDFs.

MGFs of Sums

MGF of a Sum

Theorem

If X and Y are independent, then

$$M_{X+Y}(t) = M_X(t) \times M_Y(t)$$

More generally, if $X_1, \dots, X_n$ are independent, then

$$M_{X_1 + \dots + X_n}(t) = \prod_{i=1}^n M_{X_i}(t)$$

This is one of the main reasons MGFs are useful. Since using the MGF Uniqueness theorem, the MGF characterizes the distribution in full.

Binomial MGF

A Binomial random variable can be written as

$$X = \sum_{j=1}^n I_j$$

where the I_j are independent $\text{Bern}(p)$ random variables. Since each Bernoulli MGF is

$$M_{I_j}(t) = pe^t + q$$

we get

$$M_X(t) = \prod_{j=1}^n M_{I_j}(t) = (pe^t + q)^n$$

Negative Binomial MGF

(Geom) $(X_j = 1, 2, \dots) \text{ Bin}(p)$

If

$$X \sim \text{NB}(r, p)$$

then

$$X = I_1 + \dots + I_r$$

where the I_j are independent $\text{Geom}(p)$ random variables. Thus,

$$M_X(t) = \left(\frac{pe^t}{1 - qe^t} \right)^r$$

$G_j = \# \text{ of trials needed to observe } j\text{th success after the } (j-1)\text{th success}$

Sums via MGFs

Sum of Independent Poissons

$X_i \sim \text{Pois}(\lambda_i) \Rightarrow M_{X_i}(t) = e^{\lambda_i(e^t - 1)}$

if $X_1, \dots, X_n$ are independent, then

$$M_{X_1 + \dots + X_n}(t) = \prod_{i=1}^n M_{X_i}(t) = e^{(\lambda_1 + \dots + \lambda_n)(e^t - 1)}$$

Therefore,

$$X_1 + \dots + X_n \sim \text{Pois}(\lambda_1 + \dots + \lambda_n)$$

Remark: This can be generalized to n independent Poisson r.v.s using induction.

Sum of Independent Normals

If

$$X_i \sim N(\mu_i, \sigma_i^2) \Rightarrow M_{X_i}(t) = e^{t\mu_i + \frac{1}{2}\sigma_i^2 t^2}$$

independently, then

$$M_{X_1 + \dots + X_n}(t) = \prod_{i=1}^n M_{X_i}(t) = e^{t(\mu_1 + \dots + \mu_n) + \frac{1}{2}(\sigma_1^2 + \dots + \sigma_n^2)t^2}$$

Thus,

$$X_1 + \dots + X_n \sim N(\mu_1 + \dots + \mu_n, \sigma_1^2 + \dots + \sigma_n^2)$$

Remark: This can be generalized to n independent Normal r.v.s using induction.

Distributions Closed Under Addition Under Independence

Distribution	Sum Distribution
Binomial	$X_1 \sim \text{Bin}(n_1, p), X_2 \sim \text{Bin}(n_2, p) \Rightarrow X_1 + X_2 \sim \text{Bin}(n_1 + n_2, p)$
Poisson	$X_1 \sim \text{Pois}(\lambda_1), X_2 \sim \text{Pois}(\lambda_2) \Rightarrow X_1 + X_2 \sim \text{Pois}(\lambda_1 + \lambda_2)$
Negative Binomial	$X_1 \sim \text{NB}(r_1, p), X_2 \sim \text{NB}(r_2, p) \Rightarrow X_1 + X_2 \sim \text{NB}(r_1 + r_2, p)$
Geometric	$X_1 \sim \text{Geom}(p), X_2 \sim \text{Geom}(p) \Rightarrow X_1 + X_2 \sim \text{Geom}(p)$
Normal	$X_1 \sim N(\mu_1, \sigma_1^2), X_2 \sim N(\mu_2, \sigma_2^2) \Rightarrow X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
Gamma	$X_1 \sim \text{Gamma}(\alpha_1, \lambda), X_2 \sim \text{Gamma}(\alpha_2, \lambda) \Rightarrow X_1 + X_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$
Exponential	$X_1 \sim \text{Exp}(\lambda), X_2 \sim \text{Exp}(\lambda) \Rightarrow X_1 + X_2 \sim \text{Exp}(\lambda)$

Summary

- MGFs encode moments through derivatives.
- MGFs determine distributions.
- Independent sums correspond to products of MGFs.
- MGFs simplify moment calculations dramatically.
- The Normal and Exponential MGFs are especially important.

Next lecture:

Joint distributions.

$$q(x) = a_0 + a_1(x-b) + a_2(x-b)^2 + \dots + a_n(x-b)^n$$

General def. of a polynomial:

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_{n-1}x^{n-1} + a_nx^n$$

↳ this is a (single variable) polynomial of degree n (if $a_n \neq 0$)

$$a_0, a_1, a_2, \dots, a_n \in \mathbb{R}$$

quadratic function:

$$f(x) = a \cdot x^2 + b \cdot x + c \quad a, b, c \in \mathbb{R}$$

$$a \neq 0 \Rightarrow \deg(f) = 2$$

$a = 0 \Rightarrow f$ reduces to a linear function:

$$f(x) = b \cdot x + c$$

$$p'(x) = a_1 + 2 \cdot a_2 x + 3 \cdot a_3 x^2 + \dots + (n-1) \cdot a_{n-1} x^{n-2} + n \cdot a_n x^{n-1}$$

take $\mathbb{E}[\cdot]$ from both sides

$$M_X(t) = \mathbb{E}(e^{tX})$$

provided this expectation is finite.

Using the Taylor expansion,

$$M_X(t) = \mathbb{E} \left(\sum_{k=0}^{\infty} \frac{t^k X^k}{k!} \right) = \sum_{k=0}^{\infty} \frac{t^k}{k!} \mathbb{E}(X^k)$$

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A sufficient condition is that, for some $r > 0$,

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Therefore,

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whenever the required interchange is valid. Equivalently,

$$M_X(t) = 1 + t\mathbb{E}(X) + \frac{t^2}{2!} \mathbb{E}(X^2) + \dots$$

Comparing this with the Taylor expansion

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we obtain

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Therefore,

$$X_1 + \dots + X_n \sim \text{Pois}(\lambda_1 + \dots + \lambda_n)$$

Remark: This can be generalized to n independent Poisson r.v.s using induction.

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independently, then

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Thus,

$$X_1 + \dots + X_n \sim N(\mu_1 + \dots + \mu_n, \sigma_1^2 + \dots + \sigma_n^2)$$

Remark: This can be generalized to n independent Normal r.v.s using induction.

Distributions Closed Under Addition Under Independence